GGG-P-791c
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SUPERSEDING
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August 31, 1955

FEDERAL SPECIFICATION

PULLERS, NAIL, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers nail pullers for use in extracting from wood nails of sizes up to and including 12-penny nails.

1.1.1 *Federal specification coverage.*—This Federal specification does not cover all types and classes of nail pullers which are commercially available, but only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.*—Nail pullers covered by this specification shall be of the following types and classes, as specified (see 6.1).

Type I—With standard movable jaw.
- Class 1—Box joint, with integral handguard.
- Class 2—Side joint or box joint, with attached handguard.

Type II—With handgrip on movable jaw.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking of Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific

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procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of nail pullers otherwise conforming to this specification.

3.2 **Material.**—Material used in the construction of pullers shall be as hereinafter specified. All forgings and castings shall be of uniform quality, free from blowholes, porosity, hard spots, shrinkage, cracks or other injurious defects. Strength and other essential physical properties of the forgings and castings shall be adequate throughout for the purpose intended. When materials are used which are not specifically designated they shall be entirely suitable for the purpose intended.

3.3 **Finish.**—The rammer shall be japanned or enameled. The shank and jaws shall have a natural steel finish or they may be polished or coated with japan or enamel in accordance with the manufacturer's trade practice.

3.4 **Hardness.**

3.4.1 *Jaws.*—The jaws shall be properly heat-treated to withstand abrasion or spalling and shall show not less than 45 nor more than 50 on a Rockwell C scale when tested as specified in 4.4. The hardness impression on the jaws shall be taken not less than ¼ to ¾ inch from the point.

3.4.2 *Pivot pin or bolt.*—When a shoulder is not provided on the shank or long jaw to take the impact, the pivot pin or bolt shall be heat treated to withstand abrasion or spalling and shall show not less than 45 nor more than 50 on a Rockwell C scale when tested as specified in 4.4. When a shoulder is provided on the shank or long jaw to take the impact, a carbon steel pin shall be provided without heat-treatment.

3.5 **Interchangeability.**—All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances and clearances, in order that such parts may be completely interchangeable, replaced or adjusted without requiring modification.

3.6 **Identification marking.** — Nail pullers shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.7 **Type I, with standard movable jaw.**—Type I nail pullers shall consist essentially of a shank, fixed jaw, standard movable jaw, and a rammer. The pullers shall be designed to provide sufficient space between the rammer and handguard on the shank when the rammer is in a closed position.

3.7.1 *Jaws and shank.* — The shank, the jaw which forms a part of the shank (known as the fixed jaw), and the short jaw or foot (known as the movable jaw) shall be of good quality forged steel. The jaws shall be ¼ inch, plus $\frac{1}{32}$ inch and minus $\frac{1}{16}$ inch, in width and shall be in alignment within a tolerance of $\frac{1}{32}$ inch. The jaws shall be pivoted by a joint so designed that the jaws are held open by gravity when the tool is held vertically with the jaws pointed down. The pivot pin or bolt shall be steel alloy, with a minimum diameter of ¼ inch except where a shoulder is provided on the shank or long jaw to take the impact, a carbon steel pin may be provided. The jaws shall be properly shaped to draw the nails, which have been driven into the wood with the head flush, without slipping. The shank shall be provided with a suitable handgrip not less than 3⅝ nor more than 5 inches long and shall have a suitable handguard.

3.7.2 *Rammer.*—The rammer shall be of a material that is able to withstand repeated blows of sufficient power without developing detrimental fatigue or ruptures and shall be not less than 11 nor more than 14 inches

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long. The handgrip on the rammer shall be not less than 3¾ nor more than 5 inches long. There shall be a flange not less than 1½ inches in diameter or a 1 7/16- by 1⅝-inch rectangular-shaped flange directly below the handgrip on the rammer to prevent the hand from slipping during the striking operation. The axial travel of the rammer on the shank or fixed jaw shall be not less than 5 nor more than 7 inches and shall have a smoothly guided action on the shank or fixed jaw. There shall be a positive mechanical means of preventing the separation of the rammer from the shank or fixed jaw.

3.7.3 *Weight and length.*—The length of the nail puller, when closed, shall be not less than 17 nor more than 20 inches. The weight shall be not less than 3¾ nor more than 6 pounds.

3.7.4 *Class 1, box joint, with integral hand guard.*—Class 1 nail pullers shall be of the box-joint type designed to keep the fixed jaw in proper alignment and to allow free movement of the movable jaw with a minimum of clearance. The handguard at the upper portion of the handgrip shall be integral with the shank. The class 1 nail puller shall be similar to figure 1.

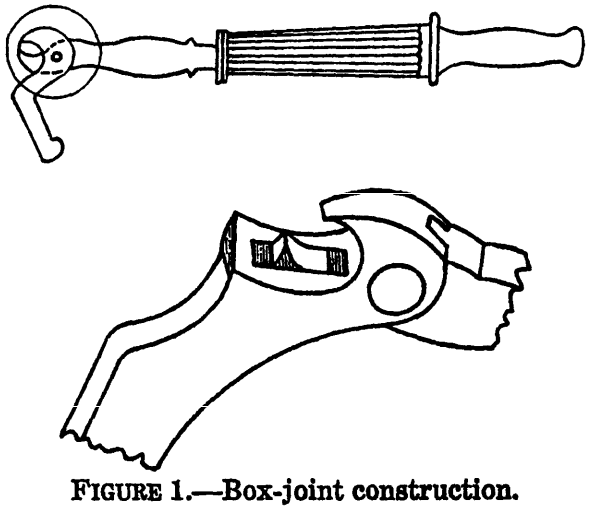


FIGURE 1.—Box-joint construction.

3.7.5 *Class 2, side joint or box joint with attached handguard.* — Class 2 nail pullers shall be either a side-joint or box-joint type having a suitable handguard at the upper portion of the handgrip on the shank. The box-joint type shall be designed to keep the fixed jaw in proper alignment and allow free movement of the movable jaw with a minimum of clearance. Class 2 nail pullers shall be similar to figure 2.

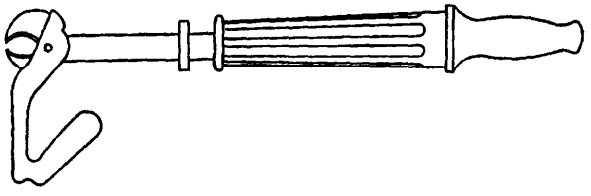


FIGURE 2.—Type I, class 2, with standard movable jaw, side joint or box joint with attached handguard.

3.8 **Type II, with handgrip on movable jaw.**—Type II nail pullers shall be similar to figure 3 and shall be in conformance with the requirements of type I, class 2 pullers (see 3.7.5) except that in lieu of a standard movable jaw, a movable jaw with integral handgrip shall be furnished.

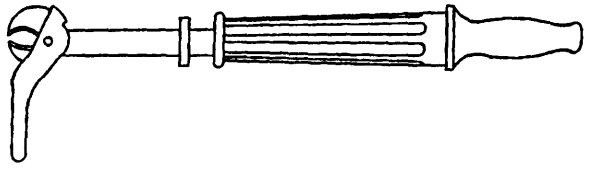


FIGURE 3.—Type II, with handgrip on movable jaw.

3.8.1 *Movable jaw with handgrip.*—The handgrip shall afford a comfortable and adequate grip and shall be not less than 3 nor more than 4 inches long.

3.9 **Workmanship.**—Workmanship shall be of the highest grade throughout and shall be equal in every respect to the best commercial practice. The pullers shall be free from rust, fins, burrs, external sharp edges, corners or surfaces and defects which may impair their serviceability or durability.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Lot.**—All nail pullers of the same type and size presented at one time shall be considered a lot for purposes of acceptance inspection.

4.3 **Sampling for lot acceptance.**

4.3.1 *Sampling for examination.*

4.3.1.1 *Sampling for intermittent procurement in isolated lots.*—A random sample of pullers shall be taken from each lot of material offered for examination with lot acceptance based on the sampling examination requirements shown in table I, in accordance with Military Standard MIL–STD–105.

TABLE I.—*Sampling for examination*
AQL (approx.) = 1.5 percent defective

Number of nail pullers in a lot	Sample size number of nail pullers to be drawn and examined (refer to Standard MIL–STD–105)
2 to 40	Inspection level III
41 to 800	Inspection level II
801 and over	Inspection level I

4.3.1.2 *Sampling for continuous production.*—Where there is continuous production for the Government and lots include at least 200 items, sampling inspection shall be conducted in accordance with Military Standard MIL–STD–105 at inspection level I and AQL equals 1.5 percent defective, the type of sampling (single, double or multiple) to be determined by the inspector.

4.3.2 *Sampling for tests.*—A random sample of nail pullers shall be taken from each lot in accordance with table II and shall be subjected to the tests specified in 4.5.

TABLE II.—*Sampling for tests*
AQL (approx.) = 4.0 percent defective

Number of nail pullers in a lot	Number of nail pullers in the sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	5	0	1
16 to 40	7	0	1
41 to 110	10	1	2
111 to 300	15	1	2
301 to 500	25	2	3
501 to 800	35	3	4
801 to 1300	50	4	5
1301 and over	75	6	7

4.4 **Examination.**—Each of the sample nail pullers selected in accordance with 4.3.1 shall be examined to verify compliance with the requirements of this specification. Examination shall be conducted as specified in table III. Any nail puller in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective nail pullers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 **Tests.**

4.5.1 *Nail-pulling test.*—Each sample nail puller shall be tested under service conditions by gripping and drawing out completely 10 or more steel-wire nails (common type), varying in size up to and including 12-penny (148 mils diameter, 3¼ inches long) which have been driven with heads flush into yellow pine or other soft wood selected by the in-

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TABLE III.—*Classification of defects in accordance with Standard MIL–STD–105*

Categories	Defects
Critical:	
1.	None defined.
Major:	
101	Type and class not as specified.
102	Material not as specified; evidence of blowholes, porosity, cracks, or breaks.
103	Incomplete, component parts missing.
104	Space between rammer and handguard (rammer closed) nonconforming.
105	Jaws not formed as specified.
106	Jaws width not within the specified tolerance.
107	Jaws not aligned within the specified tolerance.
108	Jaws inoperative; pivot pin or bolt tight; jaws fail to open when the tool is held with the jaws pointed down.
109	Pivot pin or bolt diameter less than the minimum specified.
110	Rammer length not within the specified tolerance.
111	Axial travel of rammer on shank or fixed jaw not within the specified tolerance.
112	Rammer not smoothly guided on the shank or fixed jaw; rammer sticks, binds, or evidences excessive looseness.
113	Provision not provided to prevent rammer from separation from shank or fixed jaw.
114	Rammer handgrip not provided with a flange to prevent hand from slipping.
115	Puller surface not free of rust, fins, burrs or sharp edges, not smooth.
Minor:	
201	Handgrip or shank length not within the specified tolerance.
202	Puller overall length not within the specified tolerance.
203	Finish (japanned or enameled) missing, incomplete, or not as specified.
204	Marking, manufacturer's name or trademark missing, illegible, or not permanent.
205	Weight of nail puller not within the specified tolerance.

spector. Under this test the sample nail pullers shall show no indications of weakness, such as chipping, breaking, or bending of any of the parts.

4.5.2 *Hardness test.* — Each sample nail puller shall be tested for hardness to determine conformance with 3.4.

4.5.3 *Possible test failures.*—Possible test failures are defined as follows:

(*a*) Hardness:

Hardness of jaws, pivot pin or bolt (as applicable not within the specified tolerance on the Rockwell C scale.

(*b*) Nail pulling test:

Evidence of chipping, breaking, cracks, bending or deformation when puller is subjected to the specified test; jaws fail to firmly grip the head of the nails as specified.

5. PREPARATION FOR DELIVERY

(Civil agencies.—Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation and packaging.

5.1.1 *Level A.*—Pullers shall be preserved and packaged in accordance with level A of Military Specification MIL–P–15424.

5.1.2 *Level C.*—Pullers shall be cleaned, preserved, and packaged in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Pullers packaged as specified, shall be packed in accordance with Level A of Military Specification MIL–P–15424.

5.2.2 *Level B.*—Pullers packaged as specified, shall be packed in accordance with level B of Military Specification MIL–P–15424.

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5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military marking.* — In addition to any special marking required by the contract or order, or herein, interior and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class required (see 1.2.1).
(*c*) Levels of preservation, packaging and packing required (see section 5).
(*d*) Marking, if different (see 5.3).

6.2 Transportation description. — The applicable transportation description, carload and truckload weights are as follows:

Tools, not otherwise indexed by name.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—Quartermaster Corps
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Copies of this specification may be purchased for 5 cents each.

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